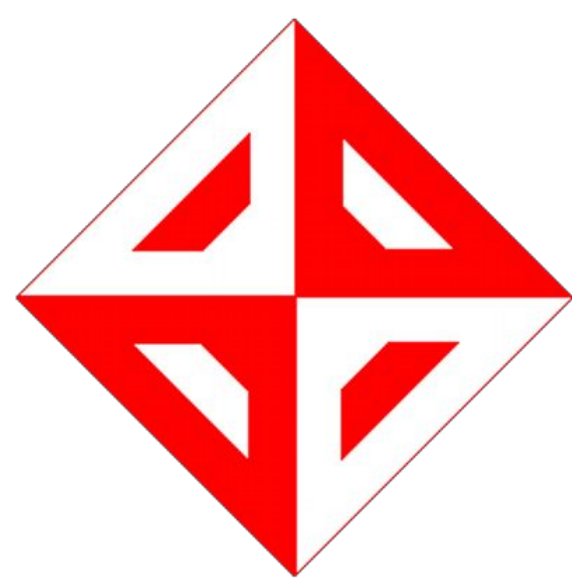
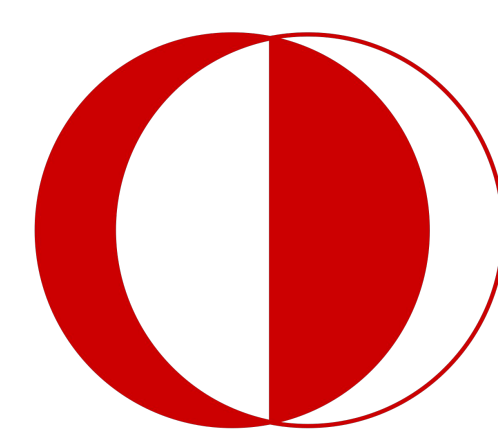


Mesh Transport Protocol (MTP) Session Management



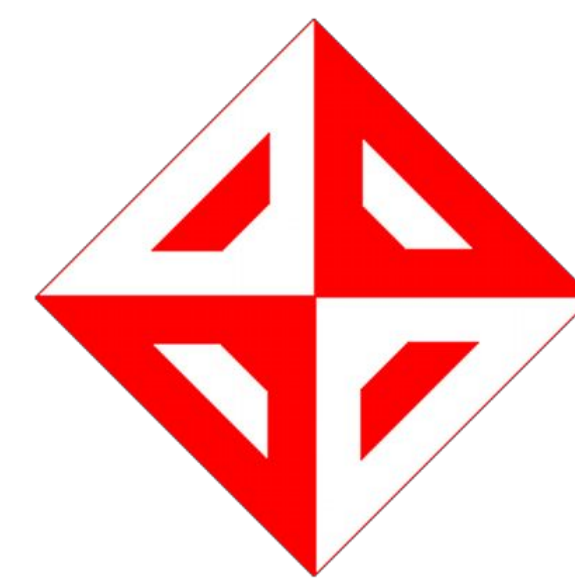
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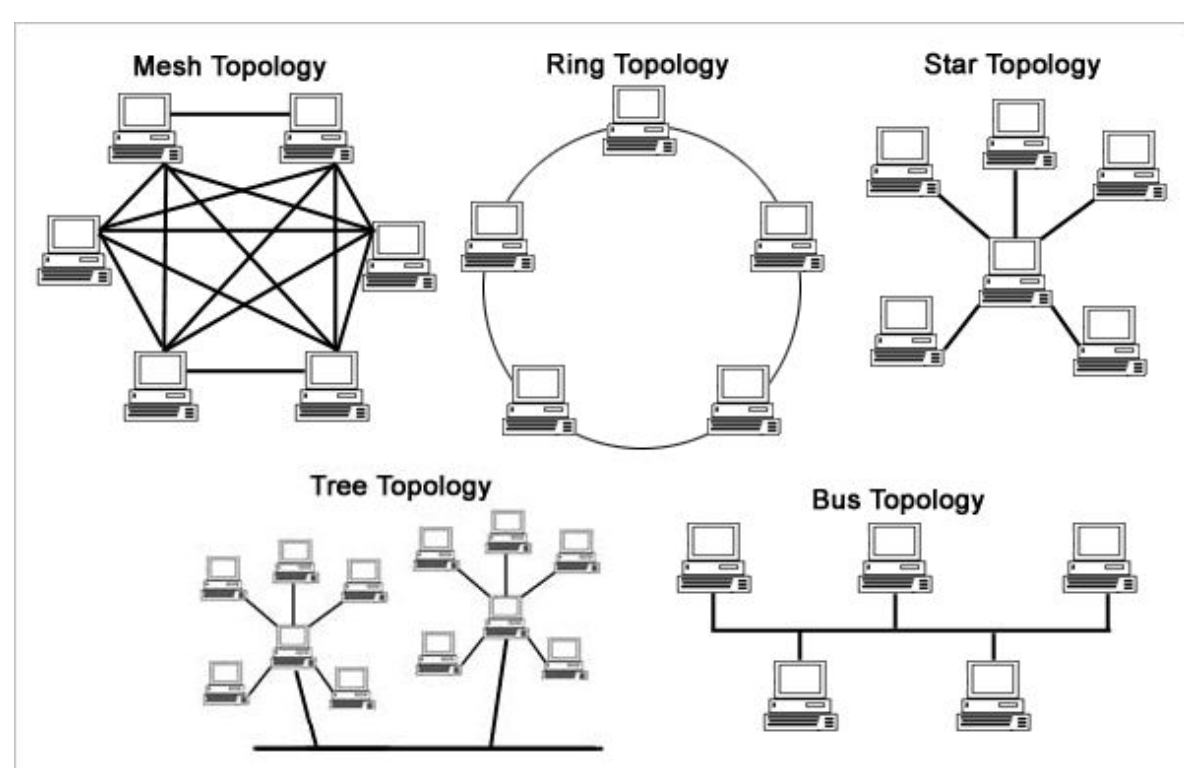
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Introduction

The modern Internet architecture faces a significant challenge known as network ossification, which restricts the flexibility and evolution of network infrastructure. This issue is primarily due to the predominance of traditional transport layer protocols like TCP and UDP, which enforce a strict one-to-one communication model. This model is increasingly misaligned with the demands of modern network usage and emerging technologies, inhibiting the development and deployment of new, advanced protocols for multicasting and broadcasting.



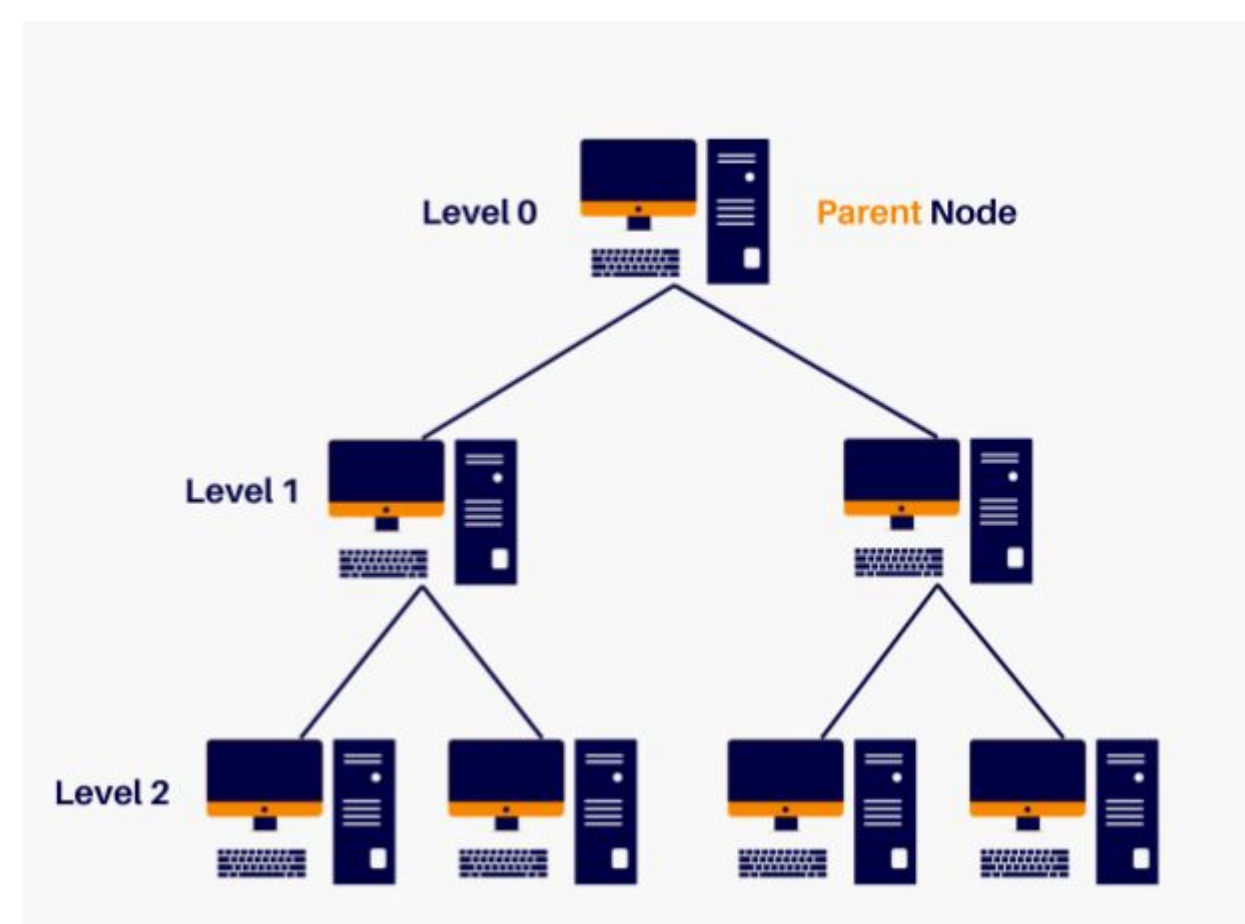
To address these limitations, we propose the Mesh Transport Protocol (MTP), built on top of the UDP protocol. MTP is designed to solve the one-to-one paradigm by enabling a more all-purpose and interconnected communication framework. Unlike traditional approaches, MTP facilitates direct, concurrent communication between a single sender and multiple clients, making it highly suitable for dynamic and interconnected network environments like Kubernetes environments.

Key Features of MTP:

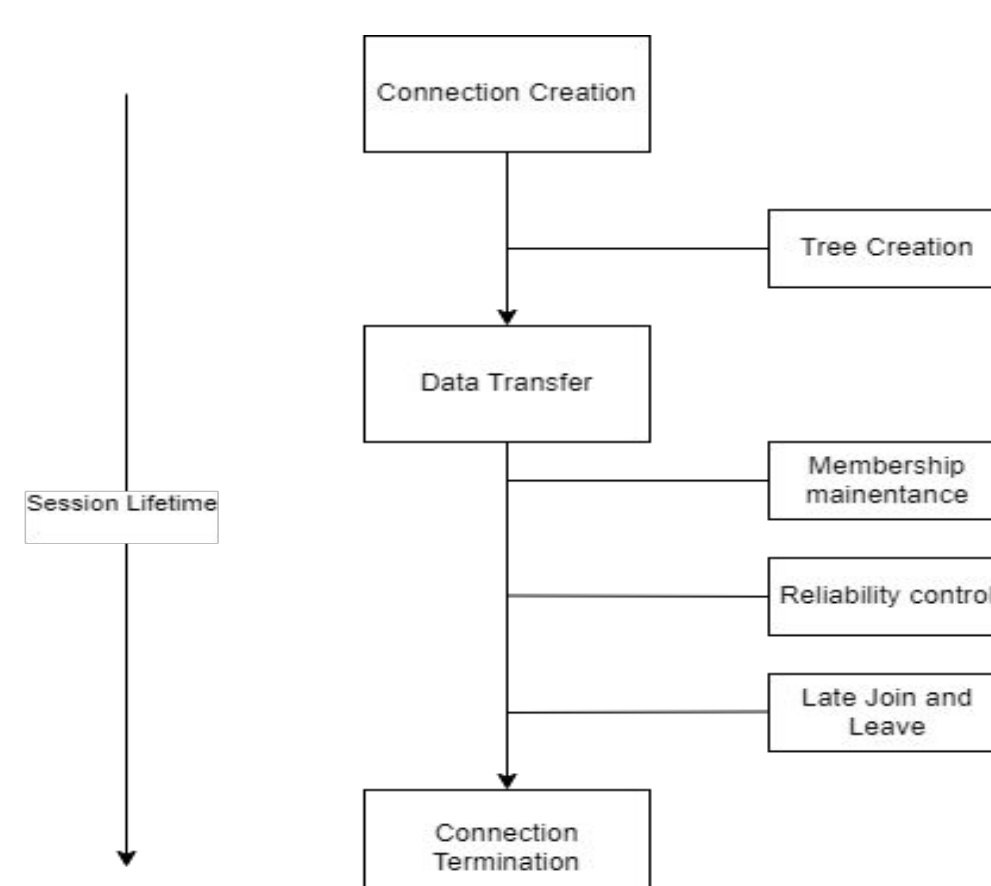
- **Dynamic Communication:** MTP allows for direct and concurrent communication between multiple nodes, provides a more flexible and resilient network structure.
- **Tree-based Participant Structure:** After extensive analysis, we selected a tree structure for its superior ability to manage dynamic network environments, offering enhanced scalability and reliability in participant management.
- **Participant Reliability and Scalability:** MTP ensures robust participant reliability and efficient scalability, maintaining data integrity across complex network topologies.
- **Ease of Integration:** MTP is designed to handle dynamic group adjustments smoothly, allowing new nodes to join tree structure seamlessly into the existing network.

Methodology

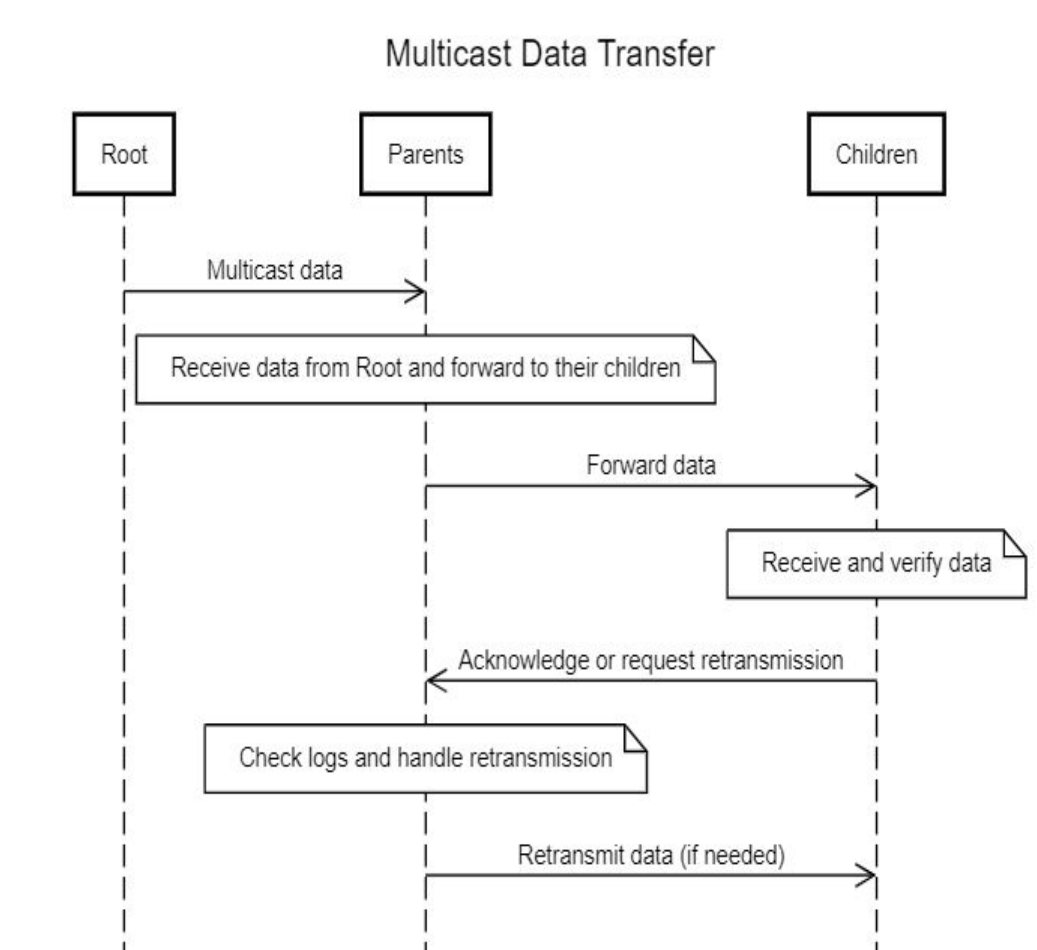
The Mesh Transport Protocol (MTP) is designed to address the limitations of traditional transport protocols by using advantage of the simplicity and speed of UDP. At its core, MTP uses a tree-based participant structure, where the sender node acts as the root and client nodes form the branches and leaves. This hierarchical organization ensures efficient data transmission and robust participant reliability, facilitating direct, concurrent communication between a single sender and multiple clients. The tree structure allows the network to dynamically adapt as nodes join or leave, maintaining optimal performance and flexibility.



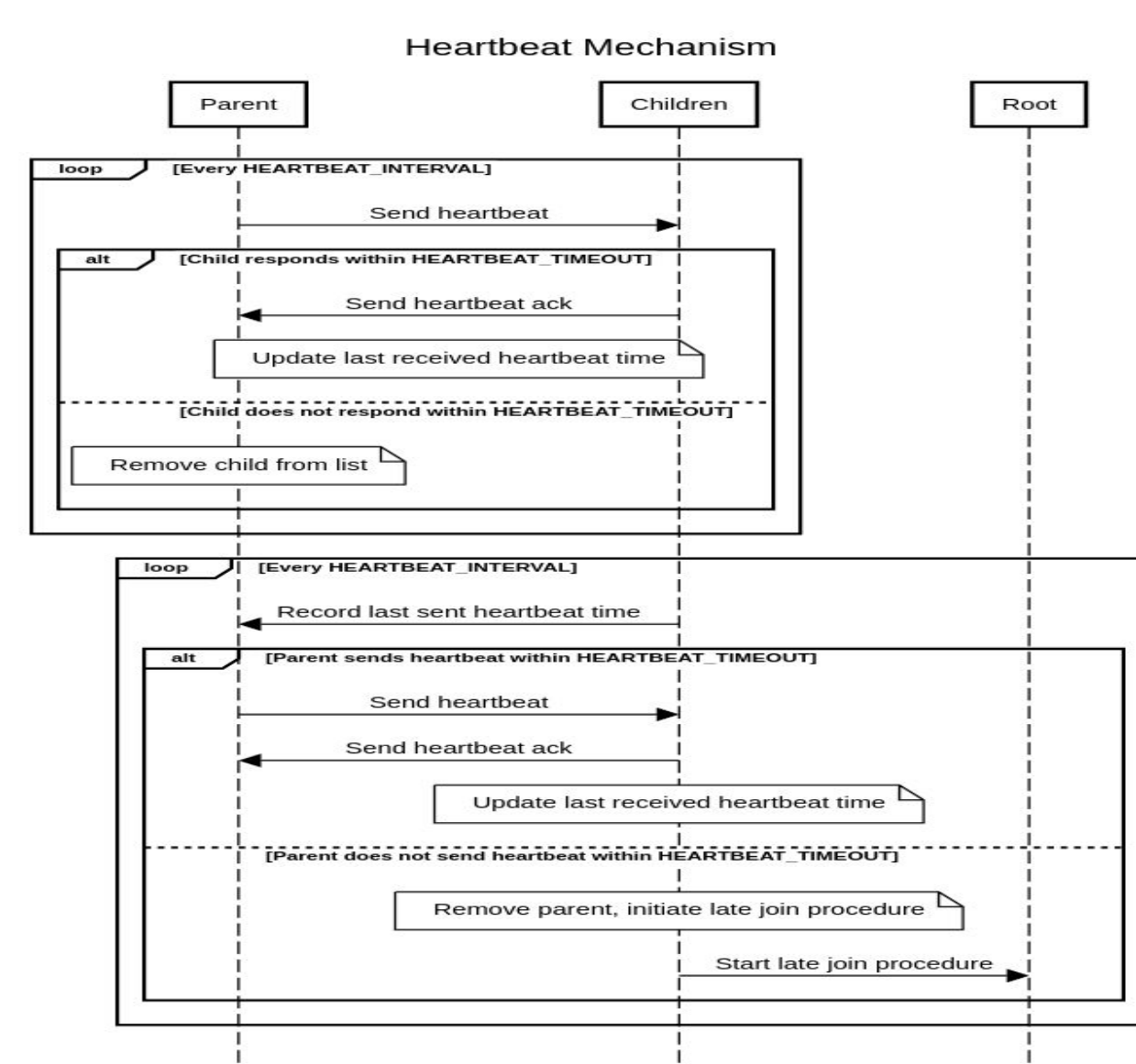
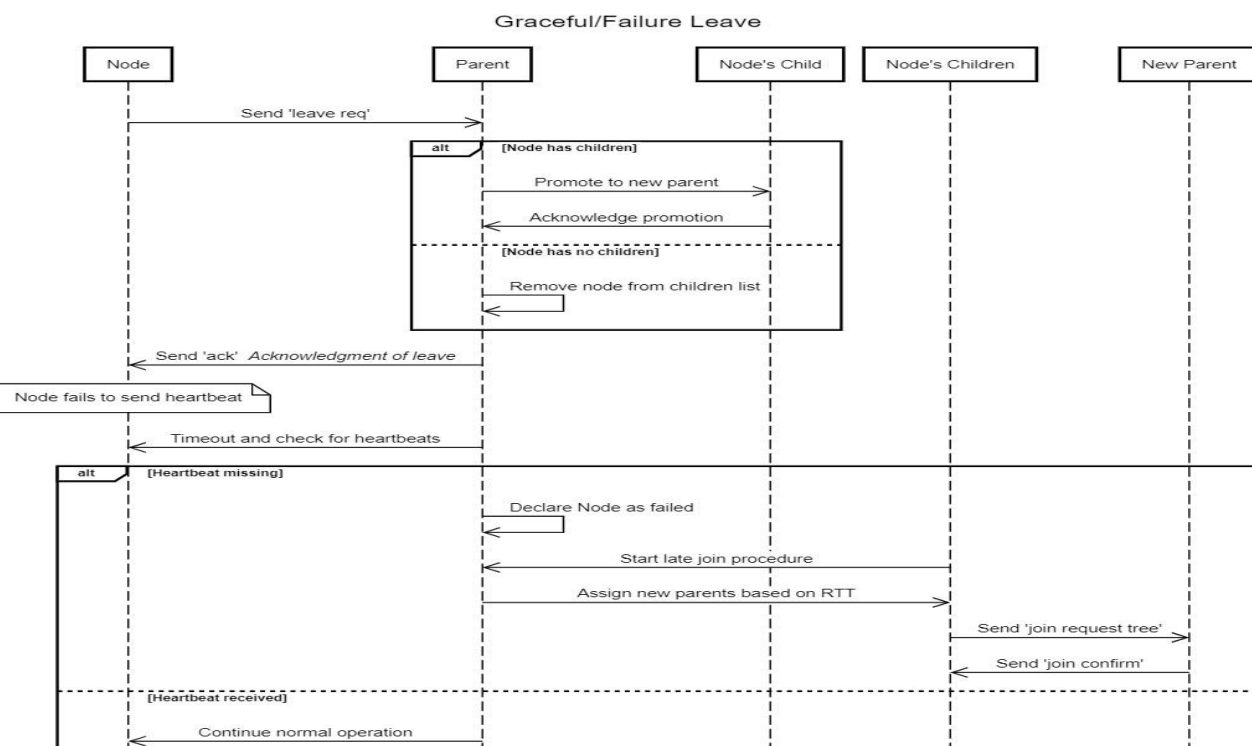
A fundamental aspect of MTP's design is its custom mechanisms for packet handling. A sequence numbering system maintains the order of packets, ensuring accurate data reconstruction at the receiver's end. Additionally, a tree-based acknowledgment mechanism enhances communication efficiency by structuring network nodes hierarchically, reducing the overhead typically associated with traditional acknowledgment processes.



MTP also supports seamless integration of new nodes, referred to as 'late joiners.' These nodes can select their parent node based on response times of parent node measurements, ensuring efficient network integration and maintaining flexibility in dynamic conditions. This feature is critical for handling dynamic group adjustments and ensuring the network's continued performance as it evolves.

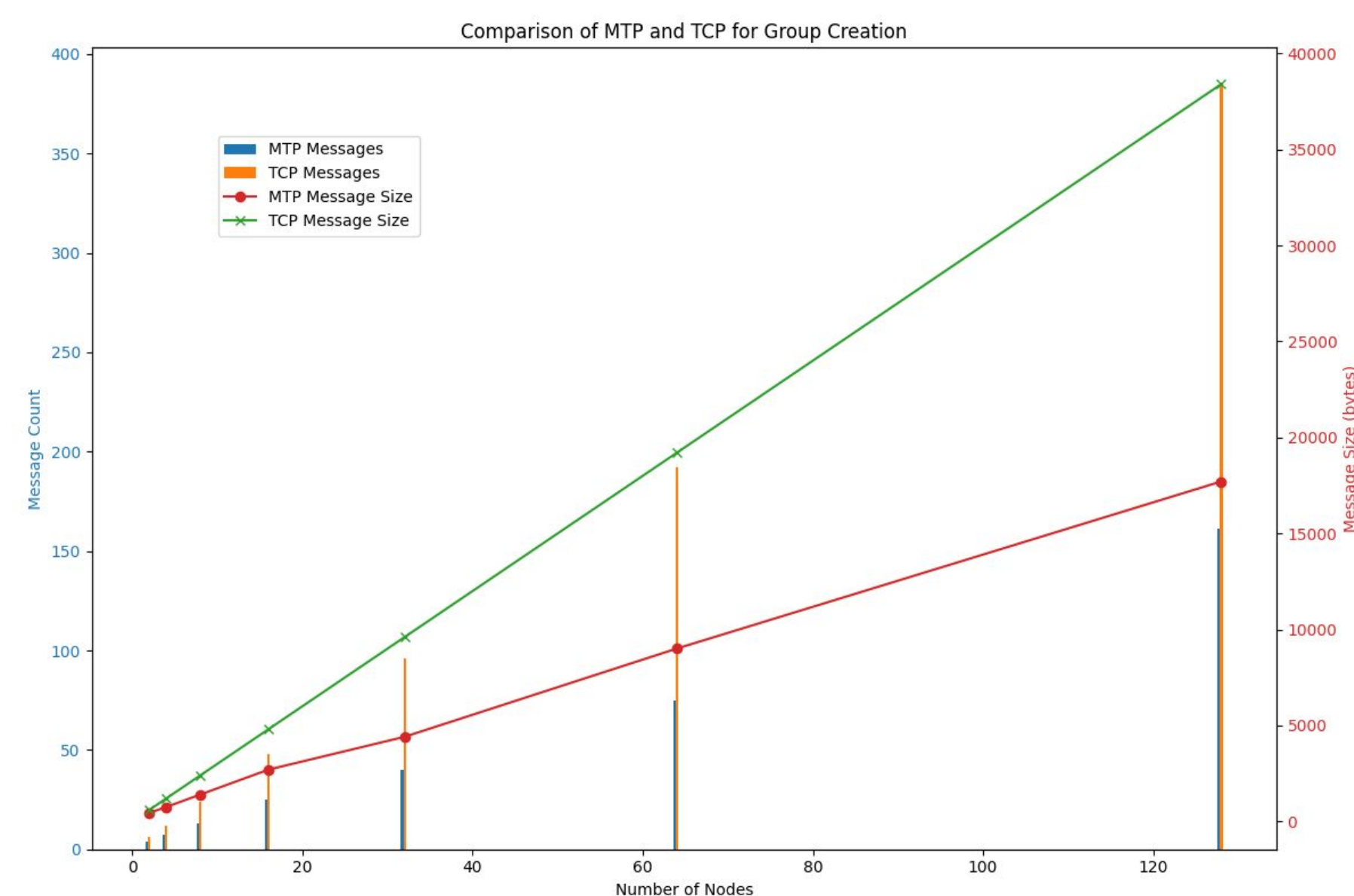


Furthermore, MTP's communication strategy involves each parent node in the tree being responsible for retransmitting lost data in response to requests from its children. This strategy ensures high data integrity across the network, minimizing data loss and supporting reliable communication.



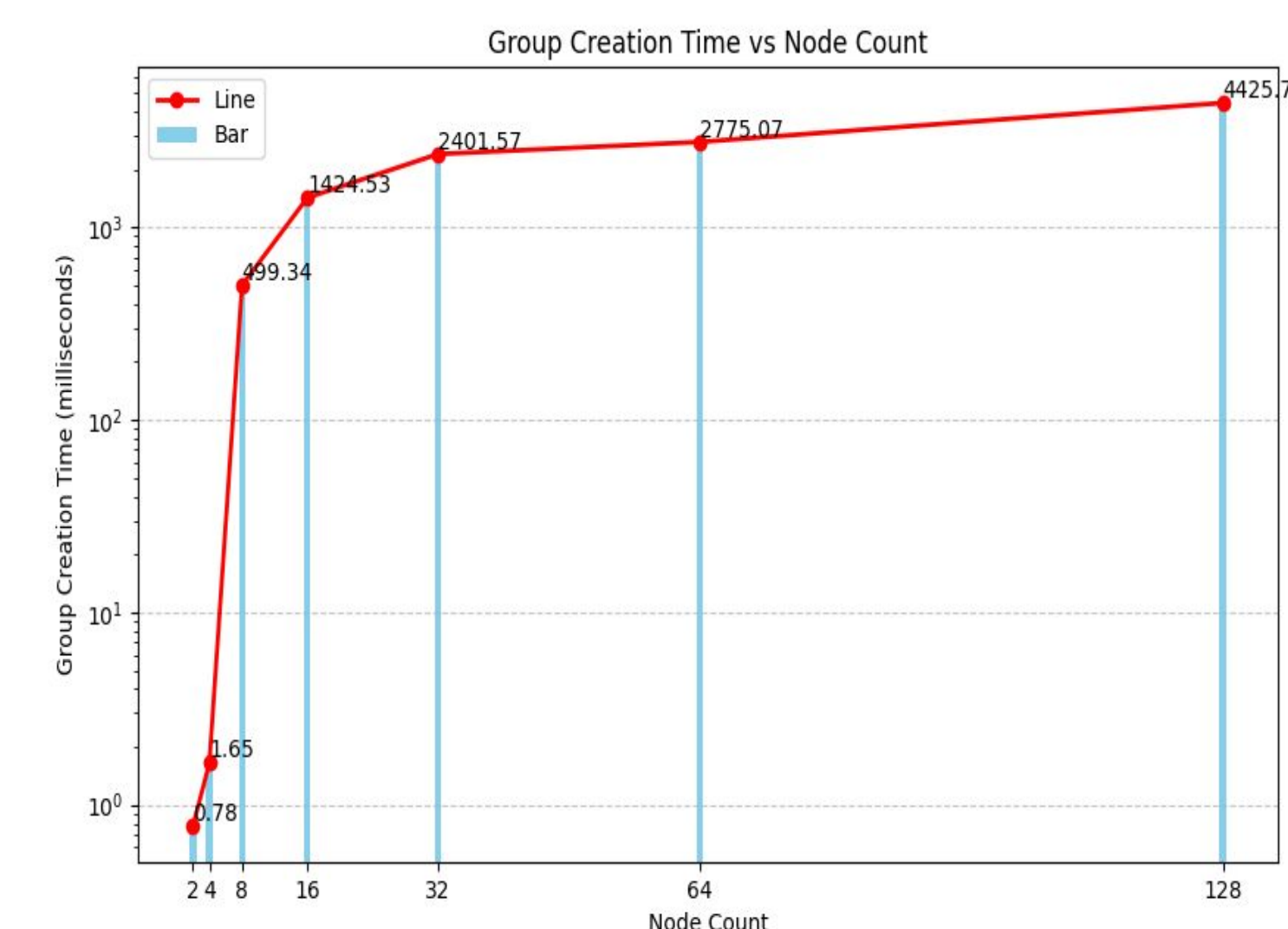
Experiment & Results

The performance of the Mesh Transport Protocol (MTP) was evaluated against the traditional TCP protocol, focusing on their efficacy in group creation within a network. The evaluation metrics included message count and message size, both of which are critical in determining the efficiency and scalability of a protocol.



Message Count: The comparison illustrated in the chart reveals that MTP consistently requires fewer messages than TCP as the number of nodes increases. For instance, when the network size reaches 128 nodes, the message count for MTP is significantly lower than that of TCP. This reduction in message count indicates MTP's superior efficiency in managing network communication, reducing overhead, and optimizing resource utilization.

Message Size: In terms of message size, MTP also demonstrates an advantage over TCP. As depicted in the chart, the message size for MTP grows at a much slower rate compared to TCP. By the time the network size reaches 128 nodes, TCP message sizes have escalated considerably, whereas MTP maintains a more manageable message size. This characteristic of MTP is crucial for ensuring that the network can scale without experiencing significant performance degradation due to large message sizes.



Scalability and Efficiency: The results highlight MTP's scalability and efficiency, particularly in environments with a growing number of nodes. The protocol's design, which incorporates a tree-based structure for packet handling and dynamic node integration, contributes to its ability to handle increased network sizes effectively. The lower message count and smaller message sizes directly translate to reduced network congestion and improved overall performance.

The Mesh Transport Protocol (MTP) represents a significant advancement in transport layer communications, specifically addressing the limitations of traditional TCP and UDP protocols. By enabling dynamic, interconnected communication between multiple nodes, MTP enhances network flexibility, scalability, and performance. Our results highlight MTP's ability to handle increasing network sizes with reduced message count and smaller message sizes, reducing congestion and improving overall efficiency. For instance, in a multicast environment, the message sender (we call them the 'root' in MTP) has fewer responsibilities compared to other structures, since parents in the tree structure help the root to establish a reliable and robust multicast communication.

Our work primarily focuses on the **session management** aspect of MTP, which includes the efficient handling of group creation, tree participant issue, join, and leave operations within the protocol. The tree-based structure and dynamic communication capabilities ensure robust packet ordering and seamless integration of new nodes, making MTP highly adaptable to evolving network conditions. This innovation supports direct, concurrent communication and offers a scalable solution for modern network demands.

In summary, MTP provides a robust, efficient, and flexible communication framework, paving the way for more adaptable and high-performing network environments in today's world where everything has to cooperate and communicate at high speeds.

Acknowledgements

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